

Transcription engines — current position and a recommendation

NoteMD · 5 October 2026 · for Mohamed Mustafa

You asked which version of Deepgram we are using, and whether to replace it with AssemblyAI Universal-3.5 Pro. The direct answer is below, followed by three findings that I think change what the next step should be.

In short

- Enhanced dictation is Azure OpenAI, not Deepgram — so please confirm which mode you were testing. Replacing Deepgram would not affect it.
- Our Deepgram setup has no keyword boosting at all and is a generation behind. Both are cheap to fix and may be most of the problem.
- Measure before migrating. Send me the audio and I will give you numbers rather than three vendors' claims.
- Hold the `gpt-transcribe` Azure deployment until the engine decision is made, so clinical validation is done once rather than twice.

1. What is running today

PATH	ENGINE	WHERE IT PROCESSES
Enhanced dictation	Azure OpenAI <code>gpt-4o-transcribe</code>	France Central
Usual recordings (consultation mode, and dictation set to "fast")	Deepgram <code>nova-2-medical</code>	EU (<code>api.eu.deepgram.com</code>)
Letter generation	Azure OpenAI <code>gpt-4o</code>	France Central

Deepgram is called with `model=nova-2-medical`, `language=en-GB`, `smart_format=true`, `punctuate=true` and `mip_opt_out=true` (the opt-out from vendor model training, applied to every request without exception).

Enhanced dictation does not use Deepgram at all. If the accuracy you were judging was in enhanced dictation, replacing Deepgram would change nothing. Before any decision, it is worth confirming which mode you were testing — the two paths share no transcription code.

2. Three findings

We are a generation behind on Deepgram

`nova-3-medical` is available, supports `en-GB`, and runs on the EU endpoint we already use. Moving to it is a one-line configuration change with the same key and the same residency position.

We are not using keyword boosting at all

This is the one I would want you to notice. Deepgram supports boosting specific vocabulary — drug names, abbreviations, dosing terms — so the model is far more likely to get them right. **We have none of it configured.** Not the older `keywords` parameter, not Nova-3's `keyterm` prompting, which accepts up to 100 clinical terms.

For medical dictation this is the cheapest accuracy lever available, and it is currently untouched. It is quite likely a real part of what you are perceiving as below standard.

AssemblyAI is not blocked on data residency

I checked this specifically, because it would have been a showstopper. It is not: Universal-3.5 Pro is available on AssemblyAI's EU endpoint, including real-time streaming, with data kept in the EU. If the accuracy case holds up, residency is not an obstacle.

3. The architectural question underneath

Two problems have come up in two days: dictation inventing content on quiet audio, and dictation missing whole passages. Both are fixed and deployed. But both are *consequences of the same design choice*, and that is worth putting in front of you rather than leaving as an implementation detail.

Enhanced dictation uses a GPT-family transcription model. These models are generative, which gives them their accuracy on difficult speech — and also means:

- **Given silence, they produce fluent invented text.** This is inherent to the model family, not a defect in our configuration. Deepgram's models do not do this; they return nothing when there is nothing. Our current safeguards detect and discard fabricated output. A different engine would remove the cause rather than filter the symptom.
- **They cannot stream.** So we cut the audio into ten-second pieces and transcribe them concurrently. Everything that went wrong last week — passages lost, sentences arriving out of order — comes from that. We now sequence, retry, detect gaps and re-transcribe the whole recording when one is found. It works, but it is machinery built to compensate for the engine being batch-only.

It also means the clinician waits 10–15 seconds for the first text to appear.

4. Options

A. Keep enhanced dictation as it is. No work. Retains the highest raw accuracy on difficult speech. Keeps the hallucination risk (mitigated, not eliminated) and the segmented architecture. Also keeps us on `gpt-4o-transcribe`, which Microsoft retires on **31 December** — mid-pilot — so a model change is required regardless.

B. Move enhanced dictation to Deepgram `nova-3-medical` with keyterm prompting. Live text instead of a 10–15 second wait. Removes the hallucination mechanism and the entire segmented pipeline. Lower cost. The open question is whether accuracy holds up — and if it does, "enhanced" and "fast" become much the same thing, which is a product decision for you.

C. Both, in sequence — my recommendation. Deepgram streams the text live during the consultation; when the clinician stops, the complete recording goes through the higher-accuracy model **once**, and that becomes the final transcript.

Option C gives immediate feedback during dictation, keeps the best available accuracy for the transcript that becomes the letter, and replaces thirty concurrent requests with one. It also sharply reduces the hallucination risk, because a single pass over a full recording containing real speech is far less prone to invention than thirty passes of which several may be near-silent.

5. What would decide it

Whether `nova-3-medical` with proper clinical keyterms closes the accuracy gap. That single measurement answers your AssemblyAI question too.

The test: the same audio through `nova-2-medical` (what you have now), `nova-3-medical` with keyterms, and `gpt-4o-transcribe`, scored on word error rate for clinical terms specifically rather than overall — a transcript can read well and still get the drug wrong.

What I need from you: a representative audio sample, ideally one of the recordings where you judged accuracy to be below standard. Your judgement is what we are measuring against, so your audio is the right benchmark.

I will include AssemblyAI in that comparison if you want it. Worth being clear that a vendor migration costs more than the integration: a new data processing agreement, a new sub-processor in the DPIA, fresh residency evidence, separate key management, and re-running clinical validation. Worth paying if Deepgram properly configured still falls short — an expensive way to discover we had simply not configured the current vendor fully.

One sequencing point. You were going to create a `gpt-transcribe` deployment on the Azure resource, which is still needed before 31 December if enhanced dictation stays on that engine. **Please hold that until this is decided.** If dictation moves to Deepgram, the deployment and its clinical validation are not needed — and validation is the expensive part, so it is worth doing once rather than twice.